

A Biologically-Plausible Framework for Mining Spatio-Temporal Sequential Patterns from Event Data Streams

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Abstract

A clear and well-documented L^AT_EX document is presented as an article formatted for publication by ACM in a conference proceedings or journal publication. Based on the “acmart” document class, this article presents and explains many of the common variations, as well as many of the formatting elements an author may use in the preparation of the documentation of their work.

CCS Concepts

• **Do Not Use This Code → Generate the Correct Terms for Your Paper**; *Generate the Correct Terms for Your Paper*; Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

Keywords

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ACM Reference Format:

Piotr S. Maciag. 2018. A Biologically-Plausible Framework for Mining Spatio-Temporal Sequential Patterns from Event Data Streams. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX)*. ACM, New York, NY, USA, 6 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Mining frequent patterns from Spatio-Temporal (ST) event data has been long attracting attention of researchers. Such patterns can naturally represent recurring phenomena in e.g., crime or environmental analysis. For example, given a dataset of crime instances, one could be interested in revealing significant sequences of crime types, instances of which occur in spatio-temporal proximity [1, 10].

In this paper, we propose a biologically-inspired framework to continuously mine sequential patterns from spatio-temporal event data. Specifically, our approach is based on spiking neural networks and the Spike-Time Dependent Plasticity (STDP) rule to modify synaptical weights in such networks. In the proposed framework, a user first provides an ST dataset of events, each of which has been assigned a location (i.e., in a two-dimensional space), an occurrence time and an event type. Subsequently, ST space of the dataset is

transformed into a 4 dimensional grid according to the user-given spatial and temporal cell resolutions. The grid is 4 dimensional as it encompasses two spatial dimensions (e.g., longitude and latitude), one temporal dimension and one dimension for event types representations. The grid is next used to construct a spiking neural network taught in an unsupervised way using STPD-like rules to reveal sequential patterns between event types.

Our approach differs significantly from previous works attempting to extract various types of patterns (such as sequential [6], cascading [14] or propagation [15]) from spatio-temporal data. While these and similar approaches are based on either (expensive) candidate pattern generation and evaluation strategies or employ graph-based mining algorithms, we deduct significant pattern from the structure of a taught spiking neural network. For example, the STS-Miner algorithm for mining spatio-temporal sequential patterns introduced by Huang et al. [6], starts the discovery process from singular event types which are recursively extended and evaluated to sequences of many event types. For the evaluation of such sequences Huang et al. proposed a new measure called sequence index. In contrast, to avoid costly and extensive iterative generation and evaluation of candidate pattern, we infer the significant sequential patterns based on strength of synaptical connections between neuron emitting spikes based on once processed input dataset.

1.1 Contributions

1.2 The Structure of the Article

The structure of the work is as follows. In Section 2, we review the related work. We split the review into two parts: (i) related to mining patterns from spatio-temporal event data, and (ii) biological foundations of spiking neural networks adapted in this work. In Section 3, we describe the proposed framework. Section 4 describes the developed learning algorithms of the neural network and the pattern inference mechanisms. In Section 5, we offer the analysis of the proposed algorithms (in terms of time and space complexity) as well as give some of its further extensions. Section 6 gives the results of experiments using real datasets. Finally, in Section 7 we discuss the achieved results and concludes the work.

2 Related Work

2.1 Mining Spatio-Temporal Sequential Patterns

In our work, we are interested in discovering STSPs from event data defined in a similar sense as by Huang et al. [6]. Specifically, they proposed a framework for mining such patterns that utilizes a depth-first recursive strategy of generating candidate patterns from already discovered significant (aka. strong) patterns. A sequential pattern was defined there as a sequence of event types

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Conference acronym 'XX, Woodstock, NY

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ACM ISBN 978-1-4503-XXXX-X/2018/06

<https://doi.org/XXXXXXX.XXXXXXX>

$\vec{s} = F_i \rightarrow \dots \rightarrow F_l$, which expresses an occurrence dependency among these types, such that, within their spatio-temporal neighbourhoods, instances of type F_1 either attract or repel occurrences of type F_2 . In turn, instances of type F_2 attract or repel occurrences of the subsequent event type, and so on, up to type F_k . The strength of the relationship between consecutive event type is quantified using the Participation Ratio (PR) measure, while the overall significance of a sequence $\vec{s}$ is evaluated using the Sequence Index (SI), defined as the minimum of all corresponding PR values. The developed pattern mining framework was implemented by the recursive STS-Miner and Slice-STS-Miner algorithms, where the latter allowed to mine pattern for large dataset that can not fit entirely into main memory. The approach presented by Huang et al. [6] was dedicated for *off-line mining* (i.e., in a non-continuous environment), where the entire dataset is available from the beginning.

Maciag et al. [12] offered an alternative approach to mine STSPs in an off-line mode. First, they proposed to discover *closed* STSPs which constitute a concise and informative representation of all STSPs. They also offered a more efficient algorithm that employs a breadth-first strategy to discover such patterns.

Mohan et al.[14] and Moosavi et al.[15] proposed alternative spatio-temporal pattern representations beyond STSPs. CSTPs model partially ordered cascades of event types, enabling richer dependency structures, while propagation patterns represent frequent embedded subtrees capturing spatio-temporal relationships between event instances. Other research has investigated mining STSPs from more complex objects, such as regions [2, 3] and trajectories [8] or mining sequential patterns from uncertain ST data [9].

2.2 Biologically-Plausible Pattern Discovery

3 Proposed Solution

3.1 Data Representation

Let $D = \{e_1, e_2, \dots\}$ denote the continuous stream of discrete event instances and $F = \{F_1, F_2, \dots, F_K\}$ the corresponding set of event types. Each event instance $e \in D$ is characterized by its spatial location ($e.location$), time of occurrence ($e.time$), and has an event type $e.F \in F$. The example of such datastream is presented by us in Fig. 1.

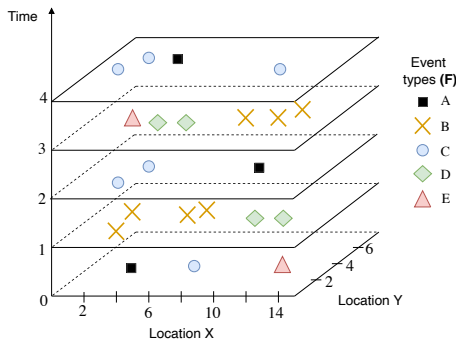


Figure 1: The example data stream of ST event instances of different types represented as a cube.

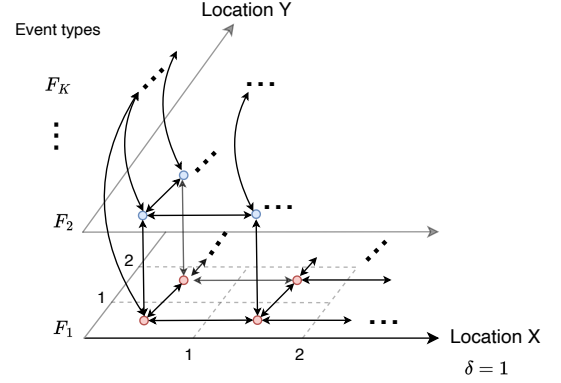


Figure 2: The architecture of the proposed solutions - at each cell neurons of different types are connected using bidirectional synapses. Neurons in the mutual spatial distance no greater than δ are also connected using bidirectional synapses.

3.2 The Architecture of the Proposed Framework

In our framework, the dataset D is used to train the proposed neural network, while the set of event types F is used to define the network's architecture. Specifically, the spatial region covered by the event instances in D is partitioned into a grid of $m_1 \cdot m_2$ rectangular cells, where m_1 and m_2 are user-specified parameters. At the center of each cell, we place a group of neurons, each corresponding to an event type in F . Within each cell, all neurons are fully connected via bidirectional synapses. That is, for any two neurons n_i and n_j in the same cell, there exist synapses $n_i \rightarrow n_j$ and $n_j \rightarrow n_i$. We also introduce bidirectional synapses between neurons belonging to different cells whose centers are within a user-defined spatial threshold.

Formally, let the spatial location of neuron n_i at the center of cell i be denoted by $n_i.loc$, and let $n_j.loc$ represent the location of neuron n_j at the center of cell j . A synaptic connection is established between n_i and n_j if and only if

$$\text{dist}(n_i.loc, n_j.loc) \leq \delta,$$

where δ is a user-specified distance threshold.

An example of the resulting network architecture is illustrated in Fig. 2. By $\mathcal{N}$ we symbolically denote the constructed neural network. Thus, the proposed network can be compared to reservoir neural networks [11] with recurrent connections between spiking neurons.

3.3 Neuronal Model

Research literature offers many different types of spiking neuronal models. To the most popular belong Leaky-Integrate-Fire (LIF) [5] in which an internal state of a spiking neuron is expressed by a membrane potential value. Such potential is accumulated with spikes incoming to the neuron. After reaching an adequate potential

threshold, the LIF neuron emits a spike resetting the membrane potential and enters a short refractory period. The literature offers many extensions to the LIF model: e.g., probabilistic [7] or non-linear LIF [13].

Specifically, each neuron corresponds to a pair consisting of an event type and a spatial cell. Let n denote a neuron assigned to event type F . Neuron n emits a spike at time t if there exists an event instance

$$e \in \mathbf{D}$$

such that

$$e.F = F$$

and the spatial location $e.loc$ belongs to the spatial cell represented by neuron n . Thus, spikes are generated directly by event occurrences in the input stream. In our approach each neuron in the network can be both:

- An input neuron emitting spikes.
- An internal neuron receiving spikes from pre-synaptic neurons.

The emitted spikes are subsequently used by the proposed Hebbian-inspired learning rule to adapt synaptic weights between neurons.

3.4 Hebbian-Inspired Unsupervised Learning

Initially, the weight of each synapse in the network is set to zero: $w_{ij} = 0$ (being also a minimal value for any weight) and there is a direct synaptic connection between n_i and n_j . During the learning process, the weights of the network are changed. Specifically, we propose to change weights of synapses according to a modified Spike-Time Dependent Plasticity (STDP) rule and modify the model proposed in [4]. The weight w_{ij} of a synapse is increased based on a timing of spikes emissions of neurons n_i, n_j . Let assume that n_j emits a spike at time t and such emissions follows emission of a spike by n_i . The change of synapse's weight between n_i, n_j is calculated according to Eq. 1.

$$\frac{dw_{ij}(t)}{dt} = \eta (A(t_j) - R(t_j)w_{ij}(t)) - \lambda w_{ij}(t), \quad (1)$$

where $A(t), R(t)$ are defined as follows:

$$\frac{dA(t)}{dt} = -\frac{1}{\tau_A}A(t), \quad \frac{dR(t)}{dt} = -\frac{1}{\tau_R}R(t). \quad (2)$$

In Eq. 1, we incorporated three main components:

- (1) $A(t)$ represents a *short-term* excitatory component of the synaptic weight. Following [4], each time the presynaptic neuron n_i fires, $A(t)$ is set to 1 and subsequently decays exponentially with time constant τ_A . Intuitively, if the post-synaptic neuron n_j fires shortly after n_i , then $A(t)$ remains close to 1, resulting in a larger increase in the synaptic weight w_{ij} . Conversely, if there is a longer delay between the firing of n_i and n_j , the value of $A(t)$ decreases, leading to a smaller weight update. The parameter η is a user-defined learning rate (e.g., 0.01).
- (2) $R(t_j)$ represents a *short-term* inhibitory and scaling component. Each time the postsynaptic neuron n_j fires, $R(t)$ is set to 1 and exponentially decays with the control parameter τ_R . This component performs a scaling role allowing to suppress the uncontrolled weight grow. In terms of ST pattern discovery, this component plays a balancing role: it allows

reduce the significance of patterns of most common event types in the data stream $\mathbf{D}$.

- (3) $\lambda w_{ij}(t)$ is a natural *long-term* weight decay component. It allows us to adapt the proposed architecture to learning from continuous data stream: over time the weights of unused synapses tend to vanish, which expresses the changing memory of the architecture and fluctuations of ST patterns in the environment. λ is a user-given decay parameter. Larger λ induces faster decay, while for smaller λ decay is slower. Usually, $\lambda \ll \tau_A, \tau_R$.

Based on Eq. 1, one can establish the maximal bound w_{max} on w_{ij} . Such a maximal bound helps to select the minimum threshold of weights in order to mine only significant ST patterns.

THEOREM 3.1. *The upper-bound w_{max} on any synaptical weight w_{ij} is equal to $\frac{\eta}{\lambda}$.*

PROOF. Let us rewrite Eq. (1) as follows:

$$\frac{dw_{ij}}{dt} = \eta A(t_j) - (\eta R(t_j) + \lambda) w_{ij}, \quad (3)$$

As stated above, we have the following assumptions $A(t) \geq 0$ and maximum of $A(t)$ is $A_{max} = 1$, $R(t) \geq 0$, $\lambda > 0$ and $\eta > 0$. Thus, the following inequality is preserved:

$$\eta R(t) + \lambda \geq \lambda. \quad (4)$$

By rewriting Eq. (3) and applying INq. (4) the following upper bound on $\frac{dw_{ij}}{dt}$ is preserved:

$$\frac{dw_{ij}}{dt} \leq \eta A_{max} - \lambda w_{ij}, \quad (5)$$

which can be further simplified by replacing A_{max} with 1:

$$\frac{dw_{ij}}{dt} \leq \eta - \lambda w_{ij}. \quad (6)$$

Assume that $w_{max} = \frac{\eta}{\lambda}$, then we have:

$$\frac{dw_{ij}}{dt} \leq \eta - \lambda w_{ij} \leq \eta - \lambda w_{max} = 0. \quad (7)$$

Thus, if at any time w_{ij} exceeds $w_{max} = \frac{\eta}{\lambda}$, its derivative $\frac{dw_{ij}}{dt}$ becomes negative, causing w_{ij} to decrease. $\square$

3.5 Pattern Definitions

So far, we have presented learning rules for individual synapses in the network. We now formally introduce the definition of an STSP and present the learning and pattern discovery algorithms.

Definition 3.2 (Spatio-Temporal Sequential Pattern, STSP). An STSP is an annotated sequence of tuples

$$\vec{s} = (F_1, loc_1) \xrightarrow{w_{12}} (F_2, loc_2) \xrightarrow{w_{23}} \dots \xrightarrow{w_{k-1,k}} (F_k, loc_k),$$

where each tuple (F_i, loc_i) corresponds to the event type and spatial location associated with neuron n_i , and w_{ij} denotes the synaptic weight between the presynaptic neuron n_i and the postsynaptic neuron n_j . An STSP $\vec{s}$ preserves the monotonic weights increase of synapses:

$$w_{12} \leq w_{23} \leq \dots \leq w_{k-1,k}.$$

Our Definition 3.2 differs from similar definitions presented in, e.g., [6], in that it explicitly accounts for the spatial location of patterns, while the strength of dependency between event types is represented by synaptic weights.

Definition 3.3 (Support). For an STSP $\vec{s}$, its support $\text{supp}(\vec{s})$ is defined as the minimum synaptic weight value of all synapses constituting $\vec{s}$.

Definition 3.4 (Significant STSP). We call an STSP $\vec{s}$ significant if its support $\text{supp}(\vec{s})$ is greater than or equal to the user-defined threshold $\theta \cdot w_{\max}$.

The definition of significance follows approaches widely used in the literature [6, 12, 14, 15]. The threshold ratio θ should satisfy $\theta \in (0, 1)$.

Let us denote the snapshot of all synapses $(n_i, n_j) \in \mathbf{W}$ that at a given time t have a synaptic weight w_{ij} greater than or equal to threshold $\theta \cdot w_{\max}$ by $\mathbf{W}$. Lemma 3.5 shows how to construct all maximal significant STSPs from $\mathbf{W}$.

LEMMA 3.5 (CONSTRUCTION OF SIGNIFICANT STSPs FROM $\mathbf{W}$). *Let $\mathbf{W}$ be the set of synapses whose weights at time t are greater than or equal to $\theta \cdot w_{\max}$, and let $\mathcal{G}_W$ be the directed graph induced by $\mathbf{W}$. Then the set of all significant STSPs at time t can be constructed by enumerating all simple directed paths in $\mathcal{G}_W$ satisfying*

$$w_{12} \leq w_{23} \leq \dots \leq w_{k-1,k},$$

and annotating every neuron n_i in each path with its corresponding event type and spatial location (F_i, loc_i) .

4 Algorithms

In this section, we present the online Algorithm 1 for mining all maximal significant STSPs. The algorithm processes the event stream $\mathbf{D}$ sequentially. For every incoming event instance $e \in \mathbf{D}$, the corresponding neuron n_t emits a spike, synaptic weights are updated according to the proposed Hebbian-inspired rule, and current maximal significant STSPs are mined and displayed to the user. The proposed algorithm takes a lazy approach: only weights that currently are in $\mathbf{W}$ and those affected by spiking of n_t are updated in when processing event e . Such approach allows us to avoid computationally heavy updates of all synaptic weights in network $\mathcal{N}$. In Section 5 we prove that such lazy approach is theoretically correct.

Algorithm 1, in addition to stream $\mathbf{D}$, requires a few user-given parameters necessary to construct neural network $\mathcal{N}$ and parameters of unsupervised learning. For each event e , the algorithm runs a sub-procedure given in Algorithm 2 that performs a local lazy update of synaptic variables and returns the set $\mathbf{M}$ of synapses needed to obtain current maximal significant STSPs. $\mathbf{M}$ consists of all synapses in $\mathbf{W}$ obtained from previous iteration of Algorithm 1, as well as of all synapses of pre or post synaptical neurons of the firing neuron n_t : S_t^{in} , S_t^{out} . For each such synapse, the traces A_{ij} and R_{ij} , as well as the synaptic weight w_{ij} , are decayed according to the elapsed time since the last update. Then, incoming synapses from neurons that fired before n_t are strengthened according to the learning rules presented in Section 3, while outgoing synapses from n_t have their activation trace set to one.

Algorithm 1: Online Significant Synapse Mining with Lazy Updates

Input: Event stream $\mathbf{D}$, grid size $m \times n$, spatial threshold δ , learning parameters $\eta, \lambda, \tau_A, \tau_R$, significance ratio θ
Output: Set of maximal significant STSPs $\mathbf{P}$

- 1 Initialize network $\mathcal{N}$ according to $\mathbf{F}$, $m \times n$, and δ ;
- 2 Set all synaptic weights $w_{ij} \leftarrow 0$;
- 3 Set all traces $A_{ij} \leftarrow 0$, $R_{ij} \leftarrow 0$;
- 4 Set $t_{ij}^{\text{last}} \leftarrow 0$ for every synapse (n_i, n_j) ;
- 5 Set $w_{\max} \leftarrow \frac{\eta}{\lambda}$; // According to Thm. 3.1.
- 6 Set $\mathbf{W} \leftarrow \emptyset$; // Initialize synapses' set $\mathbf{W}$.
- 7 **foreach** incoming event instance $e_t \in \mathbf{D}$ **do**
- 8 $c \leftarrow$ cell containing $e_t.\text{loc}$;
- 9 $n_t \leftarrow$ neuron assigned to event type $e_t.F$ in cell c ;
- 10 $t \leftarrow e_t.\text{time}$;
- 11 $\mathbf{M} \leftarrow \text{LazyWeightUpdate}(n_t, t, \mathbf{W}, \mathcal{N})$;
- 12 **foreach** synapse $(n_i, n_j) \in \mathbf{M}$ **do**
- 13 **if** $w_{ij} \geq \theta \cdot w_{\max}$ **then**
- 14 Update or Add (n_i, n_j) in $\mathbf{W}$;
- 15 **else**
- 16 Remove (n_i, n_j) from $\mathbf{W}$ if present;
- 17 $\mathbf{P} \leftarrow \text{ExtractPatternsFromSynapses}(\mathbf{W})$;
- 18 Display($\mathbf{P}$);

Algorithm 2: LazyWeightUpdate

Input: Spiking neuron n_t , current time t , significant synapse set $\mathbf{W}$, network $\mathcal{N}$, parameters $\eta, \lambda, \tau_A, \tau_R$
Output: Set $\mathbf{M}$ of updated synapses

- 1 $S_t^{\text{in}} \leftarrow \{(n_i, n_t) : (n_i, n_t) \in \mathcal{N}\}$;
- 2 $S_t^{\text{out}} \leftarrow \{(n_t, n_j) : (n_t, n_j) \in \mathcal{N}\}$;
- 3 $\mathbf{M} \leftarrow S_t^{\text{in}} \cup S_t^{\text{out}} \cup \mathbf{W}$;
- 4 **foreach** synapse $(n_i, n_j) \in \mathbf{M}$ **do**
- 5 $\Delta t \leftarrow t - t_{ij}^{\text{last}}$;
- 6 $A_{ij}(t) \leftarrow A_{ij}(t_{ij}^{\text{last}}) \exp\left(-\frac{\Delta t}{\tau_A}\right)$;
- 7 $R_{ij}(t) \leftarrow R_{ij}(t_{ij}^{\text{last}}) \exp\left(-\frac{\Delta t}{\tau_R}\right)$;
- 8 $w_{ij}(t) \leftarrow w_{ij}(t_{ij}^{\text{last}}) \exp(-\lambda \Delta t)$;
- 9 $t_{ij}^{\text{last}} \leftarrow t$;
- 10 **foreach** incoming synapse $(n_i, n_t) \in S_t^{\text{in}}$ **do**
- 11 **if** n_i fired before n_t **then**
- 12 $w_{it} \leftarrow w_{it} + \eta (A_{it}(t) - R_{it}(t)w_{it})$;
- 13 $w_{it} \leftarrow \max(0, w_{it})$;
- 14 $R_{it}(t) \leftarrow 1$;
- 15 **foreach** outgoing synapse $(n_t, n_j) \in S_t^{\text{out}}$ **do**
- 16 $A_{tj}(t) \leftarrow 1$;
- 17 **return** $\mathbf{M}$;

Algorithm 3: ExtractPatternsFromSynapses**Input:** Set of significant synapses $\mathbf{W}$ **Output:** Set of significant STSPs $\mathbf{P}$

```

1 Construct directed graph  $\mathcal{G}_W$  using synapses from  $\mathbf{W}$  as
  edges;
2  $\mathbf{P} \leftarrow \emptyset$ ;
3 foreach node  $n_s \in \mathcal{G}_W$  do
4   Perform DFS or BFS traversal from  $n_s$  maintaining a
     simple directed path
            $n_1 \rightarrow n_2 \rightarrow \dots \rightarrow n_l$ 
     such that every newly appended edge  $(n_i, n_j)$  satisfies
            $w_{prev} \leq w_{ij}$ ,
     where  $w_{prev}$  is the weight of the previously traversed
     edge;
5   For every valid path, construct STSP
            $P = (n_1.F, n_1.loc) \xrightarrow{w_{12}} (n_2.F, n_2.loc) \xrightarrow{w_{23}} \dots$ 
            $\xrightarrow{w_{l-1,l}} (n_l.F, n_l.loc)$ 
     and add  $P$  to  $\mathbf{P}$ ;
6 return  $\mathbf{P}$ ;

```

5 Analysis

5.1 Correctness of Algorithms

We become analysis with Theorem 5.1 stating that for every event $e \in \mathbf{D}$ and its occurrence time, $e.time$, Algorithm 1 provides the complete set of all maximal significant STSPs.

THEOREM 5.1. *After processing each event instance $e_t \in \mathbf{D}$, Algorithm 1 outputs the complete set $\mathbf{P}$ of significant STSPs induced by the current synaptic weights.*

PROOF. We prove the claim by induction over the processed event instances in the stream $\mathbf{D}$.

- (i) For the base case, before any event is processed, all synaptic weights are initialized to zero. Hence no synapse satisfies the significance condition $w_{ij} \geq \theta \cdot w_{max}$, the set $\mathbf{W}$ is empty, and consequently no significant STSP exists. Thus the algorithm correctly returns $\mathbf{P} = \emptyset$.
- (ii) Assume that after processing first event instances up to time t , the set $\mathbf{W}$ contains exactly all synapses whose current weights satisfy $w_{ij} \geq \theta \cdot w_{max}$, and that $\mathbf{P}$ contains all significant STSPs induced by $\mathbf{W}$. Consider the next event instance e_{t+1} . Algorithm 2 updates precisely those synapses whose weights may change due to the spike of neuron n_{t+1} or due to decay of already significant synapses in $\mathbf{W}$ of iteration t . Algorithm 1 then checks every synapse in $\mathbf{M}$ against the significance threshold, adding it to $\mathbf{W}$ if its updated weight satisfies $w_{ij} \geq \theta \cdot w_{max}$ and removing it otherwise. Therefore, after processing e_{t+1} , the set $\mathbf{W}$ again contains exactly all currently significant synapses.
- (iii) Finally, Algorithm 3 constructs the directed graph induced by $\mathbf{W}$ and enumerates all directed paths in this graph satisfying weights monotonicity. By Lemma 3.5, these paths correspond

exactly to the significant STSPs. Hence the resulting set $\mathbf{P}$ is complete after processing e_{t+1} . By induction, the theorem holds for every processed event instance in $\mathbf{D}$. $\square$

Theorem 5.1 shows the correctness of Algorithm 1 under the assumption that synapses' weights not affected by any firing of neuron n_t can only decay over time and that the algorithm provide the exact set of all maximal significant STSPs.

5.2 Time and Space Complexity of Algorithms

We skip the initialization steps of Algorithm 1 and focus on analysing complexity from the perspective of a single event instance (the main loop of Algorithm 1 and within called Algorithms 2, 3). This is especially important, since the algorithm is designed for fast processing of incoming instances of a data stream $\mathbf{D}$. Let us use the following notation:

- $N = m_1 \cdot m_2 \cdot |F|$ - total number of neurons in $\mathcal{N}$.
- E - total number of synapses in $\mathcal{N}$.
- d_{max} - maximum number of input/output synapses of any neuron $n \in \mathcal{N}$.
- $|\mathbf{W}|$ - number of currently significant synapses.
- $|\mathbf{P}|$ - number of significant patterns.
- l_{max} - maximal number of elements in any pattern.

For each event in $e_t \in \mathbf{D}$ we have:

- (1) Finding the target cell c and neuron n_t is done in a constant time $O(1)$ by indexing the spatial grid of the network.
- (2) The LazyWeightUpdate procedure operates over set $\mathbf{M}$ by updating each incoming and outgoing synapse of the firing neuron and then updating a weight of each synapse in $\mathbf{W}$. Thus, the complexity of this step is $O(d_{max} + |\mathbf{W}|)$.
- (3) Subsequently, the loop in step 12 of Algorithm 1 also scans the same set $\mathbf{M}$, thus we also have $O(d_{max} + |\mathbf{W}|)$.
- (4) Extracting patterns by Algorithm 3 is the most expensive step. First, the algorithm constructs the graph from $\mathbf{W}$ which is done in $O(|\mathbf{W}|)$. Second, the algorithm proceeds to enumerate all directed paths in $\mathcal{G}_W$ satisfying weights monotonicity constraint which requires $O(|\mathbf{P}| \cdot l_{max})$ steps. Thus, the time complexity of Algorithm 3 equals $O(|\mathbf{W}| + |\mathbf{P}| \cdot l_{max})$.

Overall, time complexity of Algorithm 1 for a single event instance is $O(d_{max} + |\mathbf{W}| + |\mathbf{P}| \cdot l_{max})$. Note that Algorithm 3 can be implemented incrementally using hash-based indices to improve its efficiency.

As for the space complexity, we need to store all synaptical weights E and additional memory is needed for storing significant synapses $\mathbf{W}$ as well as extracted patterns $\mathbf{P}$. Thus, the space complexity becomes $O(E + |\mathbf{W}| + |\mathbf{P}| \cdot l_{max})$.

6 Experiments

6.1 Datasets

6.2 Quantitative Experiments

6.3 Ablation Study

6.4 Qualitative Experiments

7 Discussion and Conclusions

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Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009